



3.0.1 ISRO-2013-34



In 8086, the jump condition for the instruction *JNBE* is?

- A. CF = 0 or ZF = 0 B. ZF = 0 and SF = 1 C. CF = 0 and ZF = 0 D. CF = 0

isro2013 8086 non-gatecse

Answer key

3.1

8085 Microprocessor (6)

3.1.1 8085 Microprocessor: ISRO CSE 2008 | Question: 39



Consider the following Assembly language program

```
MVIA 30 H
ACI 30 H
XRA A
POP H
```

After the execution of the above program, the contents of the accumulator will be

- A. 30 H B. 60 H C. 00 H D. contents of stack

isro2008 8085-microprocessor non-gatecse

Answer key

3.1.2 8085 Microprocessor: ISRO CSE 2008 | Question: 45



The TRAP is one of the interrupts available in INTEL 8085. Which one of the following statements is true of TRAP ?

- A. it is level triggered B. it is negative edge triggered
C. it is +ve edge triggered D. it is both +ve and -ve edges triggered

isro2008 co-and-architecture 8085-microprocessor non-gatecse

Answer key

3.1.3 8085 Microprocessor: ISRO CSE 2017 | Question: 20



Which interrupt in 8085 Microprocessor is unmaskable?

- a. RST 5.5 b. RST 7.5 c. TRAP d. Both (a) and (b)

isro2017 8085-microprocessor non-gatecse

Answer key

3.1.4 8085 Microprocessor: ISRO-2013-35



How many number of times the instruction sequence below will loop before coming out of the loop?

```
MOV AL, 00H
```

```
A1: INC AL
```

```
JNZ A1
```

- A. 1
C. 256

- B. 255
D. Will not come out of the loop.

isro2013 8085-microprocessor non-gatecse

Answer key

3.1.5 8085 Microprocessor: ISRO-2013-36



In 8085 microprocessor, the ISR for handling trap interrupt is at which location?

- A. $3CH$ B. $34H$ C. $74H$ D. $24H$

isro2013 8085-microprocessor non-gatecse

Answer key

3.1.6 8085 Microprocessor: ISRO2015-61



The contents of the flag register after execution of the following program by 8085 microprocessor will be

Program

```
SUB A
MVI B,(01)H
DCR B
HLT
```

- A. $(54)_H$ B. $(00)_H$ C. $(01)_H$ D. $(45)_H$

8085-microprocessor non-gatecse isro2015

Answer key

3.2 Addressing Modes (11)

3.2.1 Addressing Modes: GATE CSE 1996 | Question: 1.16, ISRO2016-42



Relative mode of addressing is most relevant to writing:

- A. Co – routines B. Position – independent code
C. Shareable code D. Interrupt Handlers

gate1996 co-and-architecture addressing-modes easy isro2016

Answer key

3.2.2 Addressing Modes: GATE CSE 2008 | Question: 33, ISRO2009-80



Which of the following is/are true of the auto-increment addressing mode?

- I. It is useful in creating self-relocating code
II. If it is included in an Instruction Set Architecture, then an additional ALU is required for effective address calculation
III. The amount of increment depends on the size of the data item accessed

- A. I only B. II only C. III only D. II and III only

gatecse-2008 addressing-modes co-and-architecture normal isro2009

Answer key

3.2.3 Addressing Modes: GATE IT 2006 | Question: 39, ISRO2009-42



Which of the following statements about relative addressing mode is FALSE?

- A. It enables reduced instruction size
- B. It allows indexing of array element with same instruction
- C. It enables easy relocation of data
- D. It enables faster address calculation than absolute addressing

gateit-2006 co-and-architecture addressing-modes normal isro2009

Answer key

3.2.4 Addressing Modes: ISRO CSE 2009 | Question: 21, UGCNET-Dec2012-II: 12



In which addressing mode, the effective address of the operand is generated by adding a constant value to the content of a register?

- A. Absolute mode
- B. Indirect mode
- C. Immediate mode
- D. Index mode

isro2009 co-and-architecture ugcnetcse-dec2012-paper2 addressing-modes

Answer key

3.2.5 Addressing Modes: ISRO CSE 2011 | Question: 5



MOV [BX], AL type of data addressing is called?

- A. register
- B. immediate
- C. register indirect
- D. register relative

isro2011 co-and-architecture addressing-modes

Answer key

3.2.6 Addressing Modes: ISRO CSE 2017 | Question: 19



The most appropriate matching for the following pairs :

- | | |
|------------------------------|----------------|
| X. Indirect Addressing | i. Loop |
| Y. Immediate Addressing | ii. Pointers |
| Z. Auto Decrement Addressing | iii. Constants |

- A. X-iii, Y-ii, Z-i
- B. X-ii, Y-iii, Z-i
- C. X-iii, Y-i, Z-ii
- D. X-ii, Y-i, Z-iii

isro2017 co-and-architecture addressing-modes match-the-following easy

Answer key

3.2.7 Addressing Modes: ISRO CSE 2020 | Question: 1



The immediate addressing mode can be used for

1. Loading internal registers with initial values
2. Perform arithmetic or logical operation on data contained in instructions

Which of the following is true?

- A. Only 1
- B. Only 2
- C. Both 1 and 2
- D. Immediate mode refers to data in cache

isro-2020 co-and-architecture normal addressing-modes

Answer key

3.2.8 Addressing Modes: ISRO CSE 2020 | Question: 15



A stack organized computer is characterised by instructions with

- A. indirect addressing
- B. direct addressing
- C. zero addressing
- D. index addressing

isro-2020 co-and-architecture addressing-modes normal

Answer key

3.2.9 Addressing Modes: ISRO CSE 2020 | Question: 3



Which of the following affects the processing power assuming they do not influence each other

1. Data bus capability
2. Address scheme
3. Clock speed

- A. 3 only
- B. 1 and 3 only
- C. 2 and 3 only
- D. 1, 2 and 3

isro-2020 co-and-architecture addressing-modes normal

Answer key

3.2.10 Addressing Modes: ISRO CSE 2020 | Question: 8



Consider a 32-bit processor which supports 70 instructions. Each instruction is 32 bit long and has 4 fields namely opcode, two register identifiers and an immediate operand of unsigned integer type. Maximum value of the immediate operand that can be supported by the processor is 8191. How many registers the processor has?

- A. 32
- B. 64
- C. 128
- D. 16

isro-2020 co-and-architecture addressing-modes normal

Answer key

3.2.11 Addressing Modes: ISRO-DEC2017-43



Consider an instruction of the type $LW\ R1, 20(R2)$ which during execution reads a 32-bit word from memory and stores it in a 32-bit register $R1$. The effective address of the memory location is obtained by adding a constant 20 and contents of $R2$. Which one best reflects the source operand?

- A. Immediate addressing
- B. Register addressing
- C. Register Indirect addressing
- D. Indexed addressing

isrodec2017 addressing-modes co-and-architecture

Answer key

3.3 Array (1)

3.3.1 Array: ISRO-DEC2017-38



A CPU has a 32 KB direct mapped cache with 128 – byte block size. Suppose A is a 2-d array of size 512×512 with elements that occupy 8 bytes each. Consider the code segment

```
for(i=0; i<512; i++)
{
    for(j=0; j<512; j++)
    {
        x += A[i][j];
    }
}
```

Assuming that array is stored in order $A[0][0], A[0][1], A[0][2] \dots$, the number of cache misses is

- A. 16384 B. 512 C. 2048 D. 1024

isrodec2017 co-and-architecture cache-memory array

Answer key 

3.4

CO and Architecture (3)

3.4.1 CO and Architecture: ISRO CSE 2009 | Question: 34



The microinstructions stored in the control memory of a processor have a width of 26 bits. Each microinstruction is divided into three fields. a micro operation field of 13 bits, a next address field (X), and a MUX select field (Y). There are 8 status bits in the inputs of the MUX. How many bits are there in the X and Y fields, and what is the size of the control memory in number of words?

- A. 10, 3, 1024 B. 8, 5, 256 C. 5, 8, 2048 D. 10, 3, 512

isro2009 co-and-architecture control-unit

Answer key 

3.4.2 CO and Architecture: ISRO CSE 2020 | Question: 2



Statements associated with registers of a CPU are given. Identify the false statement.

- A. The program counter holds the memory address of the instruction in execution
B. Only opcode is transferred to the control unit
C. An instruction in the instruction register consists of the opcode and the operand
D. The value of the program counter is incremented by 1 once its value has been read to the memory address register

isro-2020 co-and-architecture control-unit normal

Answer key 

3.4.3 CO and Architecture: ISRO CSE 2023 | Question: 23



For a given finite number of instructions to be executed, which architecture of the processor provides fastest execution?

- A. ANSA - Advanced Network system Architecture B. Super-scalar
C. ISA - Instruction Set Architecture D. All of the above

isro-cse-2023 co-and-architecture

Answer key 

3.5

Cache Memory (13)

3.5.1 Cache Memory: GATE CSE 2001 | Question: 1.7, ISRO2008-18



More than one word are put in one cache block to:

- A. exploit the temporal locality of reference in a program
B. exploit the spatial locality of reference in a program
C. reduce the miss penalty D. none of the above

gatecse-2001 co-and-architecture easy cache-memory isro2008

Answer key 

3.5.2 Cache Memory: GATE IT 2004 | Question: 12, ISRO2016-77



Consider a system with 2 level cache. Access times of Level 1 cache, Level 2 cache and main memory are 1 ns , 10 ns , and 500 ns respectively. The hit rates of Level 1 and Level 2 caches are 0.8 and 0.9, respectively. What is the average access time of the system ignoring the search time within the cache?

- A. 13.0 B. 12.8 C. 12.6 D. 12.4

gateit-2004 co-and-architecture cache-memory normal isro2016

Answer key

3.5.3 Cache Memory: ISRO CSE 2007 | Question: 14



The principal of the locality of reference justifies the use of

- A. virtual memory B. interrupts C. main memory D. cache memory

isro2007 co-and-architecture cache-memory

Answer key

3.5.4 Cache Memory: ISRO CSE 2007 | Question: 46



Consider a small 2-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently (LRU) scheme. The number of cache misses for the following sequence of block addresses is 8, 12, 0, 12, 8

- A. 2 B. 3 C. 4 D. 5

isro2007 co-and-architecture cache-memory

Answer key

3.5.5 Cache Memory: ISRO CSE 2011 | Question: 16



Consider a direct mapped cache with 64 blocks and a block size of 16 bytes. To what block number does the byte address 1206 map to

- A. does not map B. 6 C. 11 D. 54

isro2011 co-and-architecture cache-memory

Answer key

3.5.6 Cache Memory: ISRO CSE 2011 | Question: 42



The search concept used in associative memory is

- A. Parallel search B. Sequential search
C. Binary search D. Selection search

isro2011 co-and-architecture cache-memory

Answer key

3.5.7 Cache Memory: ISRO CSE 2017 | Question: 21



A cache memory needs an access time of 30 ns and main memory 150 ns , what is average access time of CPU (assume hit ratio = 80%)?

- A. 60 ns B. 30 ns C. 150 ns D. 70 ns

isro2017 co-and-architecture cache-memory

Answer key

3.5.8 Cache Memory: ISRO CSE 2018 | Question: 73



For a multi-processor architecture, in which protocol a write transaction is forwarded to only those processors that are known to possess a copy of newly altered cache line?

- a. Snoopy bus protocol
- b. Cache coherency protocol
- c. Directory based protocol
- d. None of the above

isro2018 cache-memory

Answer key

3.5.9 Cache Memory: ISRO CSE 2020 | Question: 43



Which of the following is an efficient method of cache updating?

- A. Snoopy writes
- B. Write through
- C. Write within
- D. Buffered write

isro-2020 co-and-architecture cache-memory normal

Answer key

3.5.10 Cache Memory: ISRO CSE 2020 | Question: 47



How many total bits are required for a direct-mapped cache with 128 KB of data and 1 word block size, assuming a 32-bit address and 1 word size of 4 bytes?

- A. 2 Mbits
- B. 1.7 Mbits
- C. 2.5 Mbits
- D. 1.5 Mbits

isro-2020 co-and-architecture cache-memory direct-mapping normal

Answer key

3.5.11 Cache Memory: ISRO-2013-16



How much speed do we gain by using the cache, when cache is used 80% of the time? Assume cache is faster than main memory.

- A. 5.27
- B. 2.00
- C. 4.16
- D. 6.09

isro2013 co-and-architecture cache-memory

Answer key

3.5.12 Cache Memory: ISRO-DEC2017-37



A two-way set associative cache memory unit with a capacity of 16KB is built using a block size of 8 words. The word length is 32 – bits. The physical address space is 4GB. The number of bits in the TAG, SET fields are

- A. 20,7
- B. 19,8
- C. 20,8
- D. 21,9

isrodec2017 cache-memory co-and-architecture

Answer key

3.5.13 Cache Memory: ISRO-DEC2017-42



In designing a computer's cache system, the cache block (or cache line) size is an important parameter. Which one of the following statements is correct in this context?

- A. Smaller block size incurs lower cache miss penalty.

- B. Smaller block size implies better spatial locality.
- C. Smaller block size implies smaller cache tag.
- D. Smaller block size implies lower cache hit time.

isrodec2017 co-and-architecture cache-memory

Answer key 

3.6 Cpu (2)

3.6.1 Cpu: ISRO CSE 2011 | Question: 41



If a microcomputer operates at 5 MHz with an 8-bit bus and a newer version operates at 20 MHz with a 32-bit bus, the maximum speed-up possible approximately will be

- A. 2
- B. 4
- C. 8
- D. 16

isro2011 co-and-architecture cpu

Answer key 

3.6.2 Cpu: ISRO CSE 2014 | Question: 10



The number of logical CPUs in a computer having two physical quad-core chips with hyper threading enabled is _____

- A. 1
- B. 2
- C. 8
- D. 16

co-and-architecture isro2014 cpu non-gatecse

Answer key 

3.7 DMA (1)

3.7.1 DMA: ISRO-2013-14



A processor is fetching instructions at the rate of 1 MIPS. A DMA module is used to transfer characters to RAM from a device transmitting at 9600 bps. How much time will the processor be slowed down due to DMA activity?

- A. 9.6ms
- B. 4.8ms
- C. 2.4ms
- D. 1.2ms

isro2013 dma

Answer key 

3.8 Disk (1)

3.8.1 Disk: ISRO CSE 2020 | Question: 46



A magnetic disk has 100 cylinders, each with 10 tracks of 10 sectors. If each sector contains 128 bytes, what is the maximum capacity of the disk in kilobytes?

- A. 1,280,000
- B. 1280
- C. 1250
- D. 128,000

isro-2020 co-and-architecture disk normal

Answer key 

3.9 Floating Point Representation (2)



3.9.1 Floating Point Representation: ISRO CSE 2007 | Question: 53



In the Big-Endian system, the computer stores

- A. MSB of data in the lowest memory address of data unit
- B. LSB of data in the lowest memory address of data unit
- C. MSB of data in the highest memory address of data unit
- D. LSB of data in the highest memory address of data unit

isro2007 co-and-architecture floating-point-representation little-endian-big-endian

Answer key 

3.9.2 Floating Point Representation: ISRO CSE 2020 | Question: 5



An array of 2 two byte integers is stored in big endian machine in byte addresses as shown below. What will be its storage pattern in little endian machine ?

Address	Data
0×104	78
0×103	56
0×102	34
0×101	12

A.

Address	Data
0×104	12
0×103	56
0×102	34
0×101	78

B.

Address	Data
0×104	12
0×103	34
0×102	56
0×101	78

C.

Address	Data
0×104	56
0×103	78
0×102	12
0×101	34

	Address	Data
	0×104	56
D.	0×103	12
	0×102	78
	0×101	34

isro-2020 co-and-architecture floating-point-representation normal little-endian-big-endian

Answer key 

3.10 IO Handling (6)

3.10.1 IO Handling: GATE CSE 2008 | Question: 64, ISRO2009-13



Which of the following statements about synchronous and asynchronous I/O is NOT true?

- A. An ISR is invoked on completion of I/O in synchronous I/O but not in asynchronous I/O
- B. In both synchronous and asynchronous I/O, an ISR (Interrupt Service Routine) is invoked after completion of the I/O
- C. A process making a synchronous I/O call waits until I/O is complete, but a process making an asynchronous I/O call does not wait for completion of the I/O
- D. In the case of synchronous I/O, the process waiting for the completion of I/O is woken up by the ISR that is invoked after the completion of I/O

gatecse-2008 operating-system io-handling normal isro2009

Answer key 

3.10.2 IO Handling: ISRO CSE 2008 | Question: 36



An interrupt in which the external device supplies its address as well as the interrupt requests is known as

- A. vectored interrupt
- B. maskable interrupt
- C. non maskable interrupt
- D. designated interrupt

isro2008 co-and-architecture io-handling interrupts

Answer key 

3.10.3 IO Handling: ISRO CSE 2009 | Question: 78



On receiving an interrupt from an I/O device, the CPU

- A. Halts for a predetermined time
- B. Branches off to the interrupt service routine after completion of the current instruction
- C. Branches off to the interrupt service routine immediately
- D. Hands over control of address bus and data bus to the interrupting device

isro2009 co-and-architecture io-handling

Answer key 

3.10.4 IO Handling: ISRO CSE 2011 | Question: 39



Two control signals in microprocessor which are related to Direct Memory Access (DMA) are

A. INTR & INTA
C. S0 & S1

B. RD & WR
D. HOLD & HLDA

isro2011 co-and-architecture io-handling dma

Answer key

3.10.5 IO Handling: ISRO CSE 2011 | Question: 58



In DMA transfer scheme, the transfer scheme other than burst mode is

- A. cycle technique
- B. stealing technique
- C. cycle stealing technique
- D. cycle bypass technique

isro2011 co-and-architecture io-handling dma

Answer key

3.10.6 IO Handling: ISRO CSE 2018 | Question: 65



Of the following, which best characterizes computers that use memory-mapped I/O?

- a. The computer provides special instructions for manipulating I/O ports
- b. I/O ports are placed at addresses on the bus and are accessed just like other memory locations
- c. To perform I/O operations, it is sufficient to place the data in an address register and call channel to perform the operation
- d. I/O can be performed only when memory management hardware is turned on

isro2018 co-and-architecture io-handling

Answer key

3.11 Instruction Format (5)

3.11.1 Instruction Format: ISRO CSE 2009 | Question: 79



Compared to CISC processors, RISC processors contain

- A. More register and smaller instruction set
- B. larger instruction set
- C. less registers and smaller instruction set
- D. more transistor elements

isro2009 co-and-architecture instruction-format

Answer key

3.11.2 Instruction Format: ISRO CSE 2016 | Question: 24



In which class of Flynn's taxonomy, Von Neumann architecture belongs to?

- A. SISD
- B. SIMD
- C. MIMD
- D. MISD

co-and-architecture isro2016 instruction-format

Answer key

3.11.3 Instruction Format: ISRO CSE 2018 | Question: 31



A byte addressable computer has a memory capacity of $2^m KB$ (k bytes) and can perform 2^n operations. An instruction involving 3 operands and one operator needs maximum of:

- a. $3m$ bits
- b. $3m + n$ bits
- c. $m + n$ bits
- d. none of the above

isro2018 co-and-architecture instruction-format

Answer key

3.11.4 Instruction Format: ISRO CSE 2018 | Question: 6



A data driven machine is one that executes an instruction if the needed data is available. The physical ordering of the code listing does not dictate the course of execution. Consider the following pseudo-code:

- | | |
|-----------------------------------|--|
| A. Multiply E by 0.5 to get F | B. Add A and B to get E |
| C. Add B with 0.5 to get D | D. Add E and F to get G |
| E. Add A with 10.5 to get C | Assume A, B, C are already assigned values and |
- the desired output is G . Which of the following sequence of execution is valid?
- | | |
|------------------|------------------|
| a. B, C, D, A, E | b. C, B, E, A, D |
| c. A, B, C, D, E | d. E, D, C, B, A |

isro2018 co-and-architecture instruction-format

Answer key

3.11.5 Instruction Format: ISRO CSE 2020 | Question: 49



One instruction tries to write an operand before it is written by previous instruction. This may lead to a dependency called

- | | |
|----------------------|--------------------|
| A. True dependency | B. Anti-dependency |
| C. Output dependency | D. Control Hazard |

isro-2020 co-and-architecture instruction-format normal

Answer key

3.12 Interrupts (3)

3.12.1 Interrupts: ISRO CSE 2008 | Question: 37



The ability to temporarily halt the CPU and use this time to send information on buses is called

- | | |
|-------------------------|----------------------------|
| A. direct memory access | B. vectoring the interrupt |
| C. polling | D. cycle stealing |

isro2008 co-and-architecture interrupts

Answer key

3.12.2 Interrupts: ISRO CSE 2009 | Question: 22



A certain microprocessor requires 4.5 microseconds to respond to an interrupt. Assuming that the three interrupts I_1, I_2 and I_3 require the following execution time after the interrupt is recognized:

- i. I_1 requires 25 microseconds
- ii. I_2 requires 35 microseconds
- iii. I_3 requires 20 microseconds

I_1 has the highest priority and I_3 has the lowest. What is the possible range of time for I_3 to be executed assuming that it may or may not occur simultaneously with other interrupts?

- | | |
|---|---|
| A. 24.5 microseconds to 39.5 microseconds | B. 24.5 microseconds to 93.5 microseconds |
| C. 4.5 microseconds to 24.5 microseconds | D. 29.5 microseconds to 93.5 microseconds |

isro2009 co-and-architecture interrupts

Answer key

3.12.3 Interrupts: ISRO CSE 2023 | Question: 32



Which of the following is false about interrupts?

- A. Interrupts can be triggered by a hardware or by a software
- B. Hardware interrupts may be triggered by sending a signal to CPU through a system bus
- C. Software interrupts may be triggered by executing system calls
- D. Trap is a hardware generated interrupt

isro-cse-2023 co-and-architecture interrupts

Answer key

3.13

Machine Instruction (3)

3.13.1 Machine Instruction: GATE CSE 2006 | Question: 09, ISRO2009-35



A CPU has 24-bit instructions. A program starts at address 300 (in decimal). Which one of the following is a legal program counter (all values in decimal)?

- A. 400
- B. 500
- C. 600
- D. 700

gatecse-2006 co-and-architecture machine-instruction easy isro2009

Answer key

3.13.2 Machine Instruction: ISRO CSE 2008 | Question: 42



Which of the following architecture is/are not suitable for realising SIMD?

- A. Vector processor
- B. Array processor
- C. Von Neumann
- D. All of the above

isro2008 co-and-architecture machine-instruction

Answer key

3.13.3 Machine Instruction: ISRO CSE 2020 | Question: 16



A computer which issues instructions in order, has only 2 registers and 3 opcodes ADD, SUB and MOV. Consider 2 different implementations of the following basic block :

Case 1	Case 2
$t1 = a + b;$	$t2 = c + d;$
$t2 = c + d;$	$t3 = e - t2;$
$t3 = e - t2;$	$t1 = a + b;$
$t4 = t1 - t2;$	$t4 = t1 - t2;$

Assume that all operands are initially in memory. Final value of computation also has to reside in memory. Which one is better in terms of memory accesses and by how many MOV instructions?

- A. Case 2,2
- B. Case 2,3
- C. Case 1,2
- D. Case 1,3

isro-2020 co-and-architecture machine-instruction normal

Answer key

3.14

Memory Interfacing (8)

3.14.1 Memory Interfacing: ISRO CSE 2007 | Question: 37



In comparison with static RAM memory, the dynamic Ram memory has

- A. lower bit density and higher power consumption
- B. higher bit density and higher power consumption
- C. lower bit density and lower power consumption
- D. higher bit density and lower power consumption

isro2007 co-and-architecture memory-interfacing

Answer key

3.14.2 Memory Interfacing: ISRO CSE 2009 | Question: 23



The process of organizing the memory into two banks to allow 8-and 16-bit data operation is called

- A. Bank switching
- B. Indexed mapping
- C. Two-way memory interleaving
- D. Memory segmentation

isro2009 co-and-architecture memory-interfacing

Answer key

3.14.3 Memory Interfacing: ISRO CSE 2011 | Question: 54



Number of chips (128×8 RAM) needed to provide a memory capacity of 2048 bytes

- A. 2
- B. 4
- C. 8
- D. 16

isro2011 co-and-architecture memory-interfacing

Answer key

3.14.4 Memory Interfacing: ISRO CSE 2014 | Question: 1



Consider a 33 MHz cpu based system. What is the number of wait states required if it is interfaced with a 60 ns memory? Assume a maximum of 10 ns delay for additional circuitry like buffering and decoding.

- A. 0
- B. 1
- C. 2
- D. 3

co-and-architecture isro2014 memory-interfacing

Answer key

3.14.5 Memory Interfacing: ISRO CSE 2014 | Question: 17



If each address space represents one byte of storage space, how many address lines are needed to access RAM chips arranged in a 4×6 array, where each chip is $8K \times 4$ bits?

- A. 13
- B. 15
- C. 16
- D. 17

co-and-architecture memory-interfacing isro2014

Answer key

3.14.6 Memory Interfacing: ISRO CSE 2014 | Question: 40



Assume that 16-bit CPU is trying to access a double word stating at an odd address. How many memory operations are required to access the data?

- A. 1
- B. 2
- C. 3
- D. 4

Answer key **3.14.7 Memory Interfacing: ISRO CSE 2017 | Question: 18**How many 128×8 bit RAMs are required to design $32\text{ K} \times 32$ bit RAM?

- A. 512 B. 1024 C. 128 D. 32

Answer key **3.14.8 Memory Interfacing: ISRO2015-77**In $X = (M + N \times O)/(P \times Q)$, how many one-address instructions are required to evaluate it?

- A. 4 B. 6 C. 8 D. 10

Answer key **3.15 Memory Management (1)****3.15.1 Memory Management: ISRO2015-62**

The minimum time delay between the initiation of two independent memory operations is called

- A. Access time B. Cycle time C. Rotational time D. Latency time

Answer key **3.16 Microprocessors (2)****3.16.1 Microprocessors: GATE CSE 1998 | Question: 2.10, ISRO2008-17**

The address space of 8086 CPU is

- A. one Megabyte B. 256 Kilobytes
C. 1 K Megabytes D. 64 Kilobytes

Answer key **3.16.2 Microprocessors: ISRO CSE 2011 | Question: 37**Find the memory address of the next instruction executed by the microprocessor (8086), when operated in real mode for $CS=E000$ and $IP=E000$

- A. 10E00 B. 1E000 C. F000 D. 1000E

Answer key **3.17 Microprogramming (1)**

3.17.1 Microprogramming: ISRO CSE 2018 | Question: 34



Micro program is:

- a. the name of a source program in micro computers
- b. set of microinstructions that defines the individual operations in response to a machine-language instruction
- c. a primitive form of macros used in assembly language programming
- d. a very small segment of machine code

isro2018 co-and-architecture microprogramming

Answer key

3.18

Parallel Programming (1)

3.18.1 Parallel Programming: ISRO CSE 2018 | Question: 71



A particular parallel program computation requires 100 sec when executed on a single processor, if 40% of this computation is inherently sequential (i.e. will not benefit from additional processors), then theoretically best possible elapsed times of this program running with 2 and 4 processors, respectively, are:

- a. 20 sec and 10 sec
- b. 30 sec and 15 sec
- c. 50 sec and 25 sec
- d. 70 sec and 55 sec

isro2018 co-and-architecture parallel-programming

Answer key

3.19

Pipelining (10)

3.19.1 Pipelining: GATE CSE 2002 | Question: 2.6, ISRO2008-19



The performance of a pipelined processor suffers if:

- A. the pipeline stages have different delays
- B. consecutive instructions are dependent on each other
- C. the pipeline stages share hardware resources
- D. All of the above

gatecse-2002 co-and-architecture pipelining easy isro2008

Answer key

3.19.2 Pipelining: GATE CSE 2003 | Question: 10, ISRO-DEC2017-41



For a pipelined CPU with a single ALU, consider the following situations

- I. The $j + 1^{st}$ instruction uses the result of the j^{th} instruction as an operand
- II. The execution of a conditional jump instruction
- III. The j^{th} and $j + 1^{st}$ instructions require the ALU at the same time.

Which of the above can cause a hazard

- A. I and II only
- B. II and III only
- C. III only
- D. All the three

gatecse-2003 co-and-architecture pipelining normal isrodec2017

Answer key

3.19.3 Pipelining: GATE CSE 2007 | Question: 37, ISRO2009-37



Consider a pipelined processor with the following four stages:

- IF: Instruction Fetch
- ID: Instruction Decode and Operand Fetch
- EX: Execute
- WB: Write Back

The IF, ID and WB stages take one clock cycle each to complete the operation. The number of clock cycles for the EX stage depends on the instruction. The ADD and SUB instructions need 1 clock cycle and the MUL instruction needs 3 clock cycles in the EX stage. Operand forwarding is used in the pipelined processor. What is the number of clock cycles taken to complete the following sequence of instructions?

ADD	R2, R1, R0	$R2 \leftarrow R1 + R0$
MUL	R4, R3, R2	$R4 \leftarrow R3 * R2$
SUB	R6, R5, R4	$R6 \leftarrow R5 - R4$

- A. 7 B. 8 C. 10 D. 14

gatecse-2007 co-and-architecture pipelining normal isro2009

Answer key

3.19.4 Pipelining: GATE CSE 2012 | Question: 20, ISRO2016-23



Register renaming is done in pipelined processors:

- A. as an alternative to register allocation at compile time
B. for efficient access to function parameters and local variables
C. to handle certain kinds of hazards
D. as part of address translation

gatecse-2012 co-and-architecture pipelining easy isro2016

Answer key

3.19.5 Pipelining: GATE IT 2007 | Question: 6, ISRO2011-25



A processor takes 12 cycles to complete an instruction I. The corresponding pipelined processor uses 6 stages with the execution times of 3, 2, 5, 4, 6 and 2 cycles respectively. What is the asymptotic speedup assuming that a very large number of instructions are to be executed?

- A. 1.83 B. 2 C. 3 D. 6

gateit-2007 co-and-architecture pipelining normal isro2011

Answer key

3.19.6 Pipelining: ISRO CSE 2016 | Question: 19



Consider a non-pipelined processor with a clock rate of 2.5 gigahertz and average cycles per instruction of four. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume that there are no stalls in the pipeline. The speedup achieved in this pipelined processor is

- A. 3.2 B. 3.0 C. 2.2 D. 2.0

co-and-architecture pipelining isro2016

Answer key 

3.19.7 Pipelining: ISRO CSE 2020 | Question: 6



A non-pipelined CPU has 12 general purpose registers ($R_0, R_1, R_2, \dots, R_{12}$). Following operations are supported

- ADD R_a, R_b, R_r Add R_a to R_b and store the result in R_r
- MUL R_a, R_b, R_r Multiply R_a to R_b and store the result in R_r

MUL operation takes two clock cycles, ADD takes one clock cycle.

Calculate minimum number of clock cycles required to compute the value of the expression $XY + XYZ + YZ$. The variable X, Y, Z are initially available in registers R_0, R_1 and R_2 and contents of these registers must not be modified.

- A. 5 B. 6 C. 7 D. 8

isro-2020 co-and-architecture pipelining normal

Answer key 

3.19.8 Pipelining: ISRO CSE 2020 | Question: 7



Consider a 5- segment pipeline with a clock cycle time 20 ns in each sub operation. Find out the approximate speed-up ratio between pipelined and non-pipelined system to execute 100 instructions. (if an average, every five cycles, a bubble due to data hazard has to be introduced in the pipeline)

- A. 5 B. 4.03 C. 4.81 D. 4.17

isro-2020 co-and-architecture pipelining normal

Answer key 

3.19.9 Pipelining: ISRO CSE 2023 | Question: 24



An instruction pipeline can be implemented by means of

- A. LIFO buffer B. FIFO buffer C. Stack D. None of the above

isro-cse-2023 co-and-architecture pipelining

Answer key 

3.19.10 Pipelining: ISRO-2013-15



A pipeline P operating at 400 MHz has a speedup factor of 6 and operating at 70% efficiency. How many stages are there in the pipeline?

- A. 5 B. 6 C. 8 D. 9

isro2013 co-and-architecture pipelining

Answer key 

3.20

Register (2)

3.20.1 Register: ISRO CSE 2007 | Question: 08



A read bit can be read

- A. and written by CPU
- C. by peripheral and written by CPU

- B. and written by peripheral
- D. by CPU and written by the peripheral

isro2007 co-and-architecture register

Answer key

3.20.2 Register: ISRO CSE 2008 | Question: 55



The Memory Address Register

- A. is a hardware memory device which denotes the location of the current instruction being executed.
- B. is a group of electrical ckt, that performs the intent of instructions fetched from memory
- C. contains the address of the memory location that is to be read from or stored into
- D. contains a copy of the designated memory location specified by the MAR after a "read" or the new contents of the memory prior to a "write"

isro2008 co-and-architecture register

Answer key

3.21 Runtime Environment (1)

3.21.1 Runtime Environment: GATE CSE 2008 | Question: 37, ISRO2009-38



The use of multiple register windows with overlap causes a reduction in the number of memory accesses for:

- I. Function locals and parameters
- II. Register saves and restores
- III. Instruction fetches

- A. I only
- B. II only
- C. III only
- D. I, II and III

gatecse-2008 co-and-architecture normal isro2009 runtime-environment

Answer key

Answer Keys

3.0.1	C	3.1.1	C	3.1.2	D	3.1.3	C	3.1.4	C
3.1.5	D	3.1.6	A	3.2.1	B	3.2.2	C	3.2.3	D
3.2.4	D	3.2.5	C	3.2.6	B	3.2.7	C	3.2.8	C
3.2.9	D	3.2.10	B	3.2.11	D	3.3.1	A	3.4.1	A
3.4.2	A	3.4.3	B	3.5.1	B	3.5.2	C	3.5.3	D
3.5.4	C	3.5.5	C	3.5.6	A	3.5.7	A	3.5.8	C
3.5.9	A	3.5.10	D	3.5.11	C	3.5.12	B	3.5.13	A
3.6.1	B	3.6.2	D	3.7.1	D	3.8.1	C	3.9.1	A, D
3.9.2	C	3.10.1	B	3.10.2	A	3.10.3	B	3.10.4	D
3.10.5	C	3.10.6	B	3.11.1	A	3.11.2	A	3.11.3	D
3.11.4	B	3.11.5	C	3.12.1	D	3.12.2	B	3.12.3	D
3.13.1	C	3.13.2	C	3.13.3	A	3.14.1	B	3.14.2	C

3.14.3	D
3.14.8	C
3.18.1	D
3.19.5	B
3.19.10	D

3.14.4	D
3.15.1	B
3.19.1	D
3.19.6	A
3.20.1	D

3.14.5	D
3.16.1	A
3.19.2	D
3.19.7	B
3.20.2	C

3.14.6	C
3.16.2	B
3.19.3	B
3.19.8	B
3.21.1	A

3.14.7	B
3.17.1	B
3.19.4	C
3.19.9	B